4.6.2 The object data type

Often, data is hierarchical: for example, an `email` object consisting of top-level fields like `subject` as well as an inner object to hold attachments, which in turn may have properties such as the attachment file name, its type, and so on. JSON allows us to create hierarchical objects—objects wrapped in other objects. Elasticsearch has a special data type to represent a hierarchy of objects: the `object` type.

The data types for the top-level `subject` and `to` fields are `text` and `keyword`, respectively. Because an attachment is an object itself, its data type is `object`. The two properties `filename` and `filetype` in the `attachments` object can be modeled as `text` fields. With this information, we can create a mapping definition.

Listing 4.23 Schema definition with object data types

```
PUT emails
{
  "mappings": {
    "properties": {
      "to":{
        "type": "keyword"
      },
      "subject":{
        "type": "text"
      },
      "attachments":{
        "properties": {
          "filename":{
            "type":"text"
          },
          "filetype":{
            "type":"text"
          }
        }
      }
    }
  }
}
```

Top-level properties for the emails index

Inner object consisting of second-level properties

The `attachments` property is something we should draw attention to. The type of this field is `object` as it encapsulates two other fields. The two fields defined in the `attachments` inner object are no different from the `subject` and `to` fields declared at the top level, except that they are one level down.

Once the command is executed successfully, we can check the schema by invoking the `GET emails/_mapping` command.

Listing 4.24 The `GET emails/_mapping` response

```
{
  "emails" : {
    "mappings" : {
```